

SMART SHOPPING CART

A PROJECT REPORT

submitted by

ABIN SANTHOSH (IES22CS006)

ADHITH SUNIL (IES22CS007)

ADHWAITH TT (IES22CS008)

ANITTA RAPHI E (IES22CS008)

to

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of

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in

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Department of Computer Science and Engineering

IES College of Engineering

Chittilappilly, Thrissur - 680 551

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DECLARATION

We undersigned hereby declare that the Project report (“**SMART SHOPPING CART**”), submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Ms.Meethu MB**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place : Thrissur	Name of student	: ABIN SANTHOSH
Date :	Signature of student	:
	Name of student	: ADHITH SUNIL
	Signature of student	:
	Name of student	: ADHWAITH TT
	Signature of student	:
	Name of student	: ANITTA RAPHI E
	Signature of student	:

**DEPARTMENT OF IES COLLEGE OF ENGINEERING,
CHITILAPILLY**



CERTIFICATE

This is to certify that the report entitled **SMART SHOPPING CART** submitted by **ABIN SANTHOSH(IES22CS006), ADHITH SUNIL(IES22CS007), ADHWAITH TT(IES22CS008), ANITTA RAPHI E(IES22CS025)** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Guide/Supervisor

Name : Ms.Meethu MB

Signature :

Date :

Coordinator

Name : Ms.Ayesha

Signature :

Date :

(Seal of Department of Computer Science and Engineering)

Head of Department

Name : Ms.Remya PC

Signature :

Date :

Principal

Name : Dr. S Brilly Sangeetha

Signature :

Date :

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ABIN SANTHOSH(IES22CS006)

ADHITH SUNIL(IES22CS007)

ADHWAITH TT(IES22CS008)

ANITTA RAPHI E(IES22CS025)

S6 CSE (2022 Admissions)

IES College of Engineering

Thrissur -680 551 .

ABSTRACT

Traditional shopping systems that rely on barcode scanning often face inefficiencies such as long checkout queues, manual scanning errors, and delays caused by missing or damaged barcodes, which inconvenience customers and slow down the process. These challenges, along with the growing demand for streamlined retail operations and enhanced customer convenience, motivated the development of the RFID-based Smart Shopping Cart system. This innovative solution automates item detection and billing in real time by integrating RFID readers, microcontrollers, and LCD displays directly onto shopping carts, eliminating the need for manual scanning and providing customers with instant updates on product details and costs. The system addresses critical challenges including customer adaptation to new technology and ensuring reliable communication between components to improve usability. By minimizing human effort and improving retail efficiency through faster checkouts and better inventory management, the RFID Smart Shopping Cart represents a transformative advancement in retail technology, despite ongoing challenges such as network connectivity and data security, laying the groundwork for future innovations in automated shopping systems.

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SMART SHOPPING CART

CHAPTER 1

INTRODUCTION

In recent years, the retail industry has witnessed a rapid transformation driven by advances in technology and changing consumer expectations. Among these innovations, the Smart Shopping Cart stands out as a revolutionary tool aimed at enhancing the traditional shopping experience. A Smart Shopping Cart is an intelligent shopping aid equipped with a variety of modern technologies such as RFID tags, barcode scanners, weight sensors, touchscreens, and wireless connectivity. These features work together to automate and simplify many aspects of the shopping process.

The primary purpose of a Smart Shopping Cart is to streamline the shopping journey by enabling automatic identification and billing of products as customers add items to their carts. This eliminates the need for manual scanning at checkout counters, significantly reducing wait times and improving overall customer satisfaction. Furthermore, the system can provide shoppers with real-time information about the products in their cart, including prices, discounts, and nutritional information, making it easier for customers to make informed purchase decisions.

Beyond convenience, Smart Shopping Carts also support store management by collecting valuable data on shopping patterns and inventory levels, helping retailers optimize stock management and tailor marketing strategies. Many smart carts are integrated with mobile apps or cloud platforms, enabling features such as personalized shopping lists, promotional offers, and seamless payment options.

In summary, the Smart Shopping Cart represents a crucial step toward the digitalization of retail, combining convenience, efficiency, and personalized shopping experiences. As consumers increasingly seek faster and smarter shopping solutions, these carts have the potential to revolutionize how retail stores operate, providing benefits to both customers and retailers alike.

1.1 Overview

The Smart Shopping Cart is an advanced retail technology designed to transform the traditional shopping experience into a more efficient, convenient, and user-friendly process. Unlike conventional shopping carts, this system integrates multiple smart components such as sensors, barcode or RFID scanners, digital displays, and wireless communication modules to automate and enhance various shopping tasks.

At its core, the Smart Shopping Cart allows customers to scan or automatically detect products as they place them in the cart. This real-time item recognition facilitates instant billing updates, enabling shoppers to monitor their total expenses without waiting in long checkout lines. The cart's display or connected mobile app can provide customers with detailed product information, promotional offers, and personalized recommendations, enhancing decision-making during the shopping process.

In addition to improving the customer experience, Smart Shopping Carts also offer significant benefits to retailers. They help in inventory management by tracking product movement within the store and gathering data on shopping habits. This data supports better demand forecasting and tailored marketing strategies, ultimately boosting sales and operational efficiency.

The Smart Shopping Cart system typically consists of hardware components like scanners, sensors, microcontrollers, and user interfaces, combined with software platforms for data processing and communication. It may integrate with existing point-of-sale (POS) systems and mobile payment gateways to enable seamless checkout and payment options.

Overall, the Smart Shopping Cart embodies the shift toward smart retail solutions, aiming to reduce checkout time, improve customer satisfaction, and provide retailers with actionable insights, thus shaping the future of shopping.

1.2 Significance Of Study

The significance of a Smart Shopping Cart system is becoming increasingly evident as retail technology evolves. Here are some key points on why a Smart Shopping Cart is essential:

- **Enhancing Customer Convenience:** Smart Shopping Carts automate product scanning and billing, allowing customers to shop without repeatedly scanning items or waiting in long checkout lines, thereby improving the overall shopping experience.
- **Reducing Checkout Time:** By enabling automatic item identification and real-time billing updates, smart carts significantly reduce the time spent at checkout counters, increasing store efficiency and customer satisfaction.
- **Providing Real-Time Product Information:** These carts can display prices, discounts, nutritional facts, and product recommendations directly to the shopper, helping customers make informed decisions on the spot.
- **Improving Inventory Management:** Smart Shopping Carts track products as they are added or removed, providing retailers with valuable data on stock levels and shopping patterns to optimize inventory control and reduce out-of-stock situations.
- **Enabling Personalized Shopping Experiences:** Integration with customer profiles or mobile apps allows smart carts to offer personalized promotions, shopping lists, and product suggestions, increasing engagement and boosting sales.
- **Supporting Contactless and Secure Payments:** By linking with digital wallets or mobile payment systems, smart carts facilitate smooth and secure transactions, minimizing physical contact and enhancing payment security.
- **Reducing Operational Costs:** Automation of scanning and billing processes can reduce the need for extensive cashier staffing and speed up store operations, leading to cost savings for retailers.
- **Encouraging Eco-Friendly Practices:** Some smart carts can help reduce paper waste by replacing printed receipts and shopping lists with digital versions accessible via apps.

- **Enhancing Data Analytics for Retailers:** Data collected from smart carts offers insights into consumer behavior, peak shopping times, and product popularity, allowing retailers to tailor marketing strategies and improve store layouts.
- **Increasing Accessibility:** For elderly or differently-abled customers, smart carts can provide voice assistance, easy navigation, and other supportive features, making shopping more inclusive.
- **Supporting Integration with Smart Store Ecosystems:** Smart Shopping Carts can be part of a larger network of connected devices in smart stores, including smart shelves and checkout-free payment systems, contributing to the future of autonomous retail.

CHAPTER 2

RELATED WORKS

“Data Transmission Using RFID System on Smart Shopping Carts for Checkout Process Efficiency in Supermarkets at Indonesia” presents a Smart Shopping Cart system that employs RFID technology to enhance checkout efficiency. Implemented using Arduino Uno, Flutter, PostgreSQL, REST APIs, and React, the system allows real-time scanning of products and reduces checkout delays. The study demonstrates significant improvements in transaction time and scalability with an increasing number of items and customers. [1]

“Smart Shopping Cart Using RFID Technology” proposes an IoT-based cart system that overcomes barcode limitations by using RFID chips embedded in products. An RFID reader on the cart detects added or removed items and displays billing information in real-time. Integrated with a mobile app for login and payment, the system supports inventory management, stock monitoring, and automated billing, streamlining the shopping experience and reducing manpower. [2]

“IoT-Based Smart Shopping Cart Using RFID” integrates RFID tags, Arduino, and Bluetooth to automate item scanning and billing. Products are identified when placed in the cart, with real-time bill updates shown via a mobile app. The system also supports inventory tracking and wireless data transmission to a central server, minimizing manual intervention while offering scalability and enhanced security for retail environments. [3]

“RFID-Based Smart Shopping Cart for Smart Shopping Enhanced with IoT” addresses long queues and social distancing challenges by equipping carts with RFID readers and LCDs. As items are added or removed, the system updates the bill automatically. A Wi-Fi module enables online payment and real-time communication with a central billing unit, reducing physical contact and improving shopping efficiency. [4]

“IoT-Based Smart Shopping Cart Using RFID Technology” introduces a user-friendly cart system using RFID, Arduino, GSM, and keypads to facilitate automatic product detection and billing. The system aims to reduce wait times, enhance customer safety during health crises like COVID-19, and provide a cost-effective retail automation solution. [5]

“Research on Smart Shopping Cart” proposes a system for metropolitan malls using RFID-tagged items and carts with integrated readers and displays. Item details and total costs are updated in real time. ZigBee modules transmit billing data to a central unit, allowing billing directly within the cart. This system significantly reduces wait times and supports social distancing, enhancing customer satisfaction and operational efficiency. [6]

“Smart Shopping Cart with Automatic Billing and Theft Detection using IoT” by R. Sharma et al. proposes an advanced smart cart system integrating RFID, load sensors, and IoT modules to handle both billing and theft prevention. In this system, load cells detect discrepancies between scanned items and actual weight, which helps in identifying unscanned or stolen items. The RFID reader detects tagged products in real-time, and the billing is processed through a cloud-based platform that also updates inventory data. A mobile app interface provides users with real-time billing, payment options, and product recommendations. This research stands out for introducing a dual focus on enhancing customer convenience and securing retail transactions against loss, highlighting a comprehensive approach to smart retail automation. [7]

CHAPTER 3

SYSTEM DEVELOPMENTS

3.1 Proposed Model

The proposed Smart Shopping Cart system aims to enhance the retail shopping experience by automating the billing process and eliminating long checkout queues. It leverages RFID and IoT technologies for efficient product detection, real-time billing, and seamless customer interaction. The system architecture is designed to be scalable, user-friendly, and highly responsive.

- **Input:** RFID-tagged items placed into or removed from the cart.
- **Output:** Real-time billing details, total cost calculation, and digital payment confirmation.
- **Product Detection and Tracking:**

RFID-based product identification enables automatic detection of items added to or removed from the cart. The system updates the bill in real-time without manual scanning.
- **Billing and Display:**

Real-time calculation and display of product names, prices, and running total, allowing customers to track their expenses during shopping.
- **User Authentication and Interaction:**

Customers log in via a connected application, allowing them to view their bill, apply discount coupons, and initiate digital payment securely.
- **Inventory Synchronization:**

Product data is automatically synced with a central inventory system, ensuring real-time stock updates and assisting with backend inventory management.

- **Real-Time Data Transfer:**

Billing and purchase information is transmitted wirelessly to the central server or billing system to facilitate quick and seamless checkout.

- **Digital Payment Integration:**

Upon completing the shopping, users can pay directly through integrated digital wallets or payment gateways via the mobile/web app.

The Smart Shopping Cart system reduces checkout times, improves customer convenience, and ensures a smooth and efficient retail experience. Its real-time data processing and automation capabilities make it a transformative solution for modern supermarkets and shopping malls.

3.2 Existing System

The existing traditional shopping system relies on manual processes for product selection, billing, and inventory management. Customers physically pick items from shelves without any digital assistance. At checkout counters, cashiers scan each product's barcode individually, which often leads to long queues and increased waiting times. Billing is entirely dependent on the cashier, making it prone to human errors and delays. Payment methods primarily include cash, credit/debit cards, or mobile wallets, processed at the billing counter.

Product Selection System

Customers browse and select products physically without any electronic guidance or tracking. There is no automatic system to detect or track items placed in a cart or basket.

Billing System

Each item is scanned manually by the cashier using barcode readers at the checkout counter. Billing depends on cashier accuracy; the total is calculated in real-time but can be error-prone. Customers often face long queues due to sequential scanning and payment processing.

Payment System

Payments are made via cash, credit/debit cards, or digital wallets directly at checkout counters. Payment happens only after the billing process is complete, no integration with mobile apps.

Security System

Theft prevention primarily relies on CCTV cameras and physical store personnel monitoring. No automated detection for theft or unauthorized removal of items exists. Although this traditional system is widely used, it suffers from challenges such as slow check-out, potential human error, delayed inventory updates, and lack of personalized shopping experiences, motivating the need for more advanced, automated solutions.

CHAPTER 4

REQUIREMENT SPECIFICATION

4.1 Hardware Requirements

Component	Specification
RFID Reader	MFRC522 RFID Module
RFID Tags	Passive 13.56 MHz RFID Tags
Microcontroller	Arduino Uno / Arduino Mega 2560
Display Unit	16x2 LCD Display with I2C Interface
Power Supply	9V Rechargeable Battery or Power Bank

Table 4.1: Hardware Requirements

The hardware requirements for the Smart Shopping Cart system consist of an MFRC522 RFID Reader and 13.56 MHz RFID Tags to uniquely identify products. An Arduino Uno or Arduino Mega serves as the microcontroller for interfacing and processing. A 16x2 LCD display with I2C module shows item and billing information. Power is supplied by a 9V rechargeable battery or compatible power bank to ensure portability.

4.2 Software Requirements

Software	Specification
Development Environment	Arduino IDE
Programming Language	Embedded C / Arduino Language
Libraries Used	MFRC522, LiquidCrystal_I2C, Wire

Table 4.2: Software Requirements

The software development for the Smart Shopping Cart system is carried out using the Arduino IDE. Programming is done using Embedded C or Arduino's built-in language. Key libraries include 'MFRC522' for RFID communication.

4.3 Functional Requirements

Requirement	Description
Product Identification	Detect and identify products using RFID tags.
Cart Management	Add or remove items automatically as they are placed in or removed from the cart.
Real-time Billing	Maintain a running bill and display total cost on the LCD screen.
User Authentication	Allow users to log in via RFID cards or mobile apps.
Payment Integration	Enable seamless payment through app, UPI, or RFID checkout.
Inventory Update	Update stock in the backend system when items are added or removed from the cart.
Theft Detection	Trigger alerts if an item is taken without scanning or removed improperly.
Display Interface	Show item name, price, and total bill in real-time on the LCD.
Alert Notifications	Notify users or store staff in case of unauthorized actions or errors.

Table 4.3: Functional Requirements

The functional requirements define the Smart Shopping Cart's key features and capabilities. The system must detect and manage products using RFID tags and maintain an accurate bill in real-time. It allows user authentication, supports digital payment, and updates inventory automatically. Additionally, the system should detect theft, provide live billing on an LCD interface, and alert users or staff in case of irregularities, ensuring a seamless and secure shopping experience.

4.4 Non-Functional Requirements

The non-functional requirements outline the operational qualities of the Smart Shopping Cart system. These include responsiveness, reliability, and ease of use to ensure a smooth shopping experience. Security and compatibility ensure data safety and integration with existing systems, while maintainability and power efficiency contribute to the system's practicality and long-term sustainability.

Requirement	Description
Performance	The system should detect and register RFID tags within 1 second of placing the item in the cart.
Reliability	The system should function correctly for prolonged durations without failure.
Scalability	The system must support expansion across multiple carts and integration with existing retail infrastructure.
Usability	The user interface should be simple and intuitive for customers with minimal technical knowledge.
Security	Data related to billing and inventory must be protected from unauthorized access or tampering.
Maintainability	The system should allow easy maintenance and software updates with minimal downtime.
Compatibility	The cart system should be compatible with various retail management systems and RFID standards.
Power Efficiency	The cart should consume minimal power and run efficiently on a rechargeable battery for several hours.

Table 4.4: Non-Functional Requirements

CHAPTER 5

SYSTEM LEVEL DESIGN

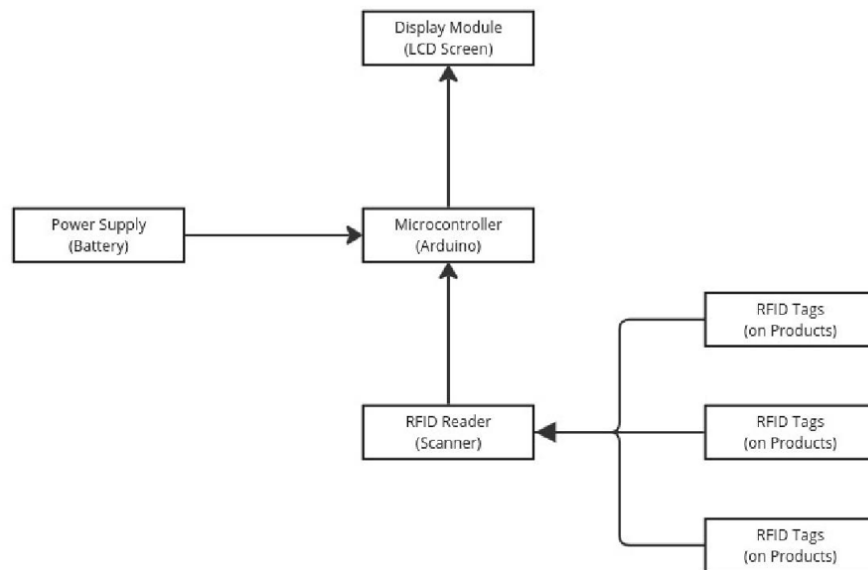


Figure 5.1: System Design

The figure illustrates a comprehensive system architecture for an RFID-based product scanning and identification system utilizing an Arduino microcontroller as the core processing unit. The entire system is powered by a battery, making it portable and suitable for various field applications without dependency on a fixed power supply. At the input stage, the RFID reader (scanner) is responsible for detecting and reading passive RFID tags that are attached to individual products. These RFID tags contain unique identification codes stored in their memory, which are wirelessly transmitted to the reader when they come into its electromagnetic field. Upon detection, the reader sends the tag data to the Arduino microcontroller, which processes the incoming information using pre-programmed logic. The microcontroller identifies the specific product based on the tag's unique ID and can be programmed to retrieve additional data such as the product name, category, or stock status. This processed information is then transmitted to an LCD display module, which serves as the output interface, allowing users to view the product details in real-time. The clear separation between power, input, processing, and output components ensures a modular

and scalable design. Such a system is particularly valuable in applications that require efficient and contactless identification of items, such as automated inventory systems, smart retail environments, library management, supply chain logistics, and access control solutions. The use of RFID technology enhances the system's efficiency by enabling fast, accurate, and non-line-of-sight scanning, which significantly reduces human effort and error compared to traditional barcode-based methods. Overall, the figure represents an intelligent, compact, and reliable design that leverages modern electronics to streamline product tracking and identification tasks. .

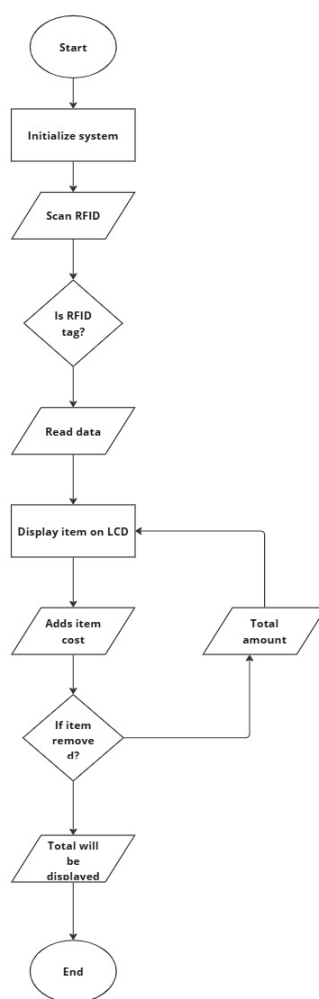


Figure 5.2: Data Flow Diagram

The Smart Shopping Cart system begins by initializing all essential components such as the microcontroller, RFID scanner, and LCD display. Once active, it waits for the

customer to scan an item's RFID tag. Upon detecting a valid RFID tag, the system reads the embedded product data, including the item name and price, and displays it on the LCD screen. The item is then added to the virtual cart, and the total amount is updated and shown. If an item is removed either by rescanning or through a designated function—the system adjusts the list of items and recalculates the total, which is again displayed. This process continues until the shopping is completed, after which the system displays the final total for checkout, marking the end of the shopping session.

5.1 System Overview

The Smart Shopping Cart system is designed to streamline the in-store shopping experience by integrating sensor technology, computer vision, and wireless communication. The system automatically detects products added or removed from the cart, tracks the bill in real-time, and enables contactless checkout. The architecture is divided into several functional modules: item detection, billing and display, user authentication, and communication with a central server.

System Overview of Smart Shopping Cart

Cart Hardware Architecture

- **Load Cell Sensors:** Detect weight variations to estimate item quantity.
- **RFID Reader:** Scans RFID tags attached to products for unique identification.
- **Camera Module:** Utilizes real-time image processing to verify item identity using computer vision.
- **Touchscreen Display:** Shows real-time cart contents, prices, and total bill.
- **Microcontroller/Processor:** ESP32 or Raspberry Pi is used to process sensor inputs and handle data communication.

User Authentication

- **Login via QR Code:** Customers scan a store-issued QR code linked to their digital wallet/account.
- **Session Initialization:** User session is created and synced with the backend server to track purchases.

Item Detection and Billing

- **RFID Scan:** Automatically detects product ID when item is placed inside the cart.
- **Weight Verification:** Cross-verifies RFID data with load cell weight to detect fraudulent items.
- **Image Recognition (optional):** Uses object detection (YOLOv5 or MobileNet) for additional verification.
- **Real-time Billing:** Price fetched from store database and total updated on-screen instantly.

Checkout Process

- **Contactless Payment:** Upon completion, the user taps 'Checkout' on the screen to pay via stored wallet or NFC.
- **E-Receipt Generation:** A digital receipt is sent to the user's registered email or app.
- **Exit Gate Sync:** Exit gate validates completed transactions via QR or RFID tag scan.

Backend Communication

- **Cloud Sync:** Cart updates product details, user data, and billing to a centralized cloud server using WiFi.

- **Inventory Management:** Product removals from shelves update real-time inventory records.
- **Analytics Dashboard:** Store managers can monitor cart usage, shopping behavior, and low-stock alerts.

Performance Optimization

- **Edge Processing:** Basic tasks such as weight checking and RFID reading are handled locally to reduce latency.
- **Lightweight CNN Models:** MobileNetV2 is used for on-device object recognition.
- **Efficient Power Usage:** Power management protocols suspend non-essential modules during inactivity.
- **Batch Data Upload:** Non-critical logs are transmitted in bulk to reduce network congestion.

Future Enhancements

- Integration with AR-based product recommendations.
- Voice command interface for accessibility.
- Dynamic price updating during promotional periods.

5.2 Deployment And Frontend

A. Web Application (Frontend)

A user-friendly web interface allows users to log in, view real-time cart activity, and complete transactions. The frontend will be built using **React.js**, ensuring an interactive and responsive user experience. Customers can view itemized billing, product details, and checkout options, while store managers can access inventory data and analytics through a secure dashboard..

B. Backend

The backend will be developed using **Django**, which integrates with embedded systems on the cart and cloud services. It will manage user authentication, product database, cart synchronization, billing computation, and transaction records. The backend will also provide REST APIs consumed by both the smart cart and frontend applicatio.

C. User Feedback

The Smart Shopping Cart system is designed to provide transparent and informative feedback to users throughout the shopping journey. Immediate updates and visual cues are displayed on the cart's touchscreen interface as well as the linked mobile/web application..

CHAPTER 6

MODULE DESCRIPTION

The Smart Shopping Cart system is composed of multiple functional modules, each dedicated to a specific aspect of intelligent retail operations. These modules work together to provide seamless, automated shopping experiences, from item detection to billing and user interaction. The following sections describe the different modules and their functionalities in detail.

6.1 User Interface Module

This module serves as the main interaction point between the user and the Smart Cart. It provides a responsive touchscreen display or web/mobile interface that displays the itemized list of products added or removed in real-time, shows the running total including price, tax, and applicable offers, and allows users to review their purchase history. It also offers system notifications for errors, product mismatch alerts, and confirmation messages. The checkout options and payment methods are easily accessible through this interface.

6.2 Product Detection Module

This core module enables real-time detection of products placed in or removed from the cart. It uses embedded sensors such as RFID, barcode scanners, or weight sensors, and optionally employs a camera with image processing techniques like YOLOv5 or SSD for visual detection. This module ensures accurate product identification, validates product presence with backend inventory, and detects duplicate entries, ensuring a reliable and automated tracking experience.

6.3 Billing and Pricing Module

This module manages all billing calculations and applies pricing logic as items are added to the cart. It calculates the subtotal, taxes, discounts, and total in real-time, retrieves up-to-date pricing and promotional information from the backend, and supports loyalty programs and membership-based pricing models. At checkout, it generates the final bill and passes it to the payment module for processing.

6.4 Inventory Management Module

This module ensures that the store's inventory remains synchronized with the cart activities. It updates the central inventory system whenever items are added or removed from carts, sends alerts to staff when product thresholds are low, and prevents the purchase of out-of-stock items. It enables real-time synchronization across multiple carts and supports efficient restocking and inventory planning.

6.5 File Handling and Data Sync Module

This module handles secure and efficient data storage and transmission throughout a shopping session. It temporarily stores user and cart data, manages API-based synchronization between the frontend and backend, and automatically clears temporary data after checkout. Additionally, it logs session data for reporting and debugging purposes, ensuring reliability and system integrity.

6.6 Logging and Monitoring Module

This module tracks system performance and user interactions to ensure reliability and provide analytical insights. It logs key activities such as item addition or removal and payment transactions, captures performance metrics and errors, and monitors cart behavior for anomalies or suspicious activity. The module also generates detailed system usage and health reports.

6.7 Result Visualization and Report Module

This module helps users and store managers better understand shopping activity and system performance. It displays graphical summaries of shopping sessions, generates receipts with item details, prices, time, and store location, and allows exporting this information as PDF or email. For administrators, it offers an analytics dashboard with insights into consumer behavior and system efficiency.

6.8 Security and Privacy Module

This module ensures that all user data and transaction information is handled securely. It encrypts sensitive data, uses secure APIs and HTTPS protocols for data transmission, implements strict access control measures for administrative features, and automatically purges data from inactive or completed sessions. The module aligns with privacy standards to guarantee customer trust and data integrity.

CHAPTER 7

PROJECT IMPLEMENTATION

7.1 Tools and Technologies Used

This section outlines the specific tools, programming languages, and technologies employed in the development of the Smart Shopping Cart system, highlighting their roles in enabling automated billing, product identification, and system integration.

Programming Languages

- **Arduino C/C++** – The Smart Shopping Cart logic is implemented using the Arduino programming language (a simplified version of C/C++), which is natively supported in the Arduino IDE. It is used for interfacing with hardware components such as the RFID reader and LCD display, as well as implementing real-time billing algorithms.

Libraries and Utilities

- **RFID Library** – Interfaces with the RFID reader module to read tag information.
- **LiquidCrystal** – Manages communication between the Arduino and the LCD display.
- **EEPROM** – Stores product data and supports persistent storage.

IDE and Tools

- **Arduino IDE** – Used for writing, compiling, and uploading code to the Arduino board.
- **draw.io** – Used to design block diagrams and represent system architecture.

Deployment Platform

- **Embedded Hardware Prototype** – The Smart Shopping Cart system is deployed and demonstrated on a hardware prototype using an Arduino board with connected RFID reader and LCD display.

Hardware Components

- **RFID Reader (RC522)** – Scans RFID tags on products.
- **RFID Tags** – Attached to items to enable unique identification.
- **Arduino Uno** – Serves as the main processing unit.
- **LCD Display (16x2)** – Displays scanned item details and total cost.
- **Rechargeable Battery Pack** – Powers the cart's electronics.

7.2 Algorithm Details

This section describes the logic that powers the Smart Shopping Cart's item detection and billing mechanism.

Input Details

The system reads inputs via RFID tags attached to products. When a tagged item is brought near the RFID reader, the tag's UID is captured. This UID is mapped to a product name and price in the system's internal database.

Algorithm Logic

The billing algorithm operates as follows:

- **Tag Reading:** When an RFID tag is scanned, the system extracts the UID.

- **Lookup and Pricing:** The UID is matched to a predefined product entry (e.g., name and price).
- **Display:** Product name and price are shown on the LCD.
- **Billing:** The price is added to a running total maintained by the Arduino.

Blocking and Error Handling

To prevent duplicate billing and incorrect scans:

- The system tracks scanned UIDs in a list to avoid repeated addition.
- Errors or unknown tags display a warning on the LCD.

Output Details

A display of the scanned item and its cost. A continuously updated total bill amount. The billing system does not rely on machine learning. Instead, it uses simple rule-based logic coded directly into the Arduino: If UID is unknown → show error. The system provides Immediate feedback through the display. On-the-spot billing and cart total. Reduced checkout delays due to pre-scanned items Though not implemented in the basic version, the system could be extended to Send data wirelessly to a central server via WiFi or GSM modules. The Smart Shopping Cart system leverages RFID technology to automate retail billing. Through real-time detection, display, and billing logic implemented in Arduino, the solution minimizes human error and improves customer experience.


```

41: }
42:
43: void loop() {
44:   // Prompt for product details
45:   String productName, productPrice, productQuantity;
46:
47:   Serial.println("Enter Product Name:");
48:   while (!Serial.available()) {} // wait for input
49:   productName = Serial.readString();
50:   productName.trim(); // Remove extra spaces
51:
52:   Serial.println("Enter Product Price:");
53:   while (!Serial.available()) {} // wait for input
54:   productPrice = Serial.readString();
55:   productPrice.trim(); // Remove extra spaces
56:
57:   Serial.println("Enter Product Quantity:");
58:   while (!Serial.available()) {} // wait for input
59:   productQuantity = Serial.readString();
60:   productQuantity.trim(); // Remove extra spaces
61:
62:   Serial.println("Now scan the RFID tag...");
63:
64:   // Wait for RFID tag
65:   while (!rfid.PICC_ShowCardPresent() || !rfid.PICC_ReadCardSerial()) {}
66:
67:   Serial.println("Card detected!");
68:
69:   // Split product name into 3 blocks
70:   byte nameLocks[3][16];
71:   splitStringIntoLocks(productName, nameLocks);
72:
73:   // Write name data to blocks 8, 9, and 10
74:   for (int i = 0; i < 3; i++) {
75:     if (writeLock(i + 1, nameLocks[i])) {
76:       Serial.print("block ");
77:       Serial.print(i + 1);
78:       Serial.println(" written successfully");
79:     }
80:   }

```

Figure 7.4: Write Data to Tags

```

41: void loop() {
42:   // Write quantity to block 11
43:   byte quantityLock[16] = {0};
44:   productQuantity.getBytes(quantityLock, 16); // Convert quantity string to bytes
45:   if (writeLock(11, quantityLock)) {
46:     Serial.println("Quantity written successfully");
47:   } else {
48:     Serial.println("Failed to write quantity.");
49:   }
50:
51:   // Wait PICC
52:   rfid.PICC_HaltA();
53:   // Stop encryption on PCID
54:   rfid.PCD_StopCrypt();
55:   // Ready for next input
56:   Serial.println("All data written successfully!\n\n*** Welcome To Product Data Writer ***\n\n");
57: }
58:
59: bool writeLock(byte block, byte *dataBlock) {
60:   MFRC522::StatusCode status = rfid.PCD_Authenticate(MFRC522::PICC_CMD_PV_AUTH_A, block, &key, &(rfid.uid));
61:   if (status != MFRC522::STATUS_OK) {
62:     Serial.print("Authentication failed for block ");
63:     Serial.print(block);
64:     Serial.println("");
65:     Serial.println(rfid.GetStatusCodeName(status));
66:     return false;
67:   }
68:
69:   status = rfid.MIFARE_Write(block, dataBlock, 16);
70:   if (status != MFRC522::STATUS_OK) {
71:     Serial.print("Write failed for block ");
72:     Serial.print(block);
73:     Serial.println("");
74:     Serial.println(rfid.GetStatusCodeName(status));
75:     return false;
76:   }
77:   return true;
78: }

```

Figure 7.5: Write Data to Tags

7.4 Outputs

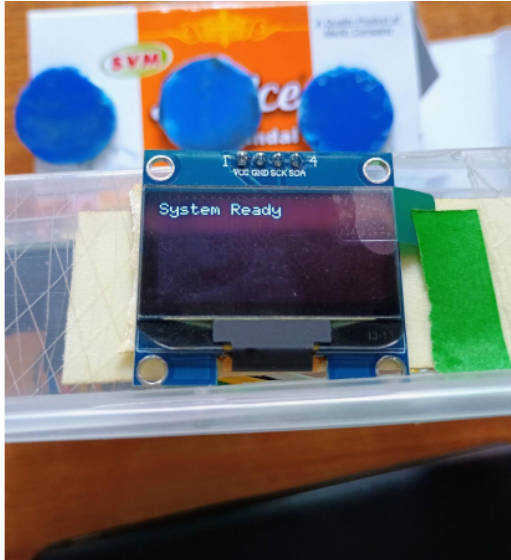


Figure 7.6: Display Module



Figure 7.7: Display Items

CHAPTER 8

TESTING METHODS

The Smart Shopping Cart system integrates hardware components (RFID reader, Arduino, LCD) and software logic to automate billing and enhance shopping convenience. The following testing methods ensure accuracy, efficiency, and reliability of the system under diverse conditions.

8.1 Unit Testing

Verifying individual modules and functions ensures correctness before integration. The core components tested include:

- `read rfid()` – Confirms accurate tag detection and data reading.
- `update bill()` – Validates proper calculation and update of total amount.
- `display lcd(message)` – Ensures real-time and correct display of item names and prices.
- `reset cart()` – Clears cart items and resets bill to default state.
- `scan item(tag_id)` – Handles duplicate item detection and item recognition logic.

Each function was tested using mock RFID tag data and verified via the Arduino Serial Monitor and LCD display.

Tools: Arduino IDE, Serial Monitor, Simulated RFID Tags

8.2 Integration Testing

Ensuring smooth interaction between hardware and software modules is crucial. The main interactions tested include:

- RFID reader → Arduino → LCD
- RFID tag detection → Item lookup → Bill update
- Reset button → Microcontroller → Bill reset

The system was tested by scanning various tags and validating that all components worked in sequence without conflict.

Tools: Arduino IDE, Serial Output Logger

8.3 Performance Testing

Evaluating the system's behavior under different operational conditions is key. Key metrics include:

- Time taken to detect and process an RFID tag.
- Bill update latency after consecutive item scans.
- LCD display refresh speed.
- System stability during long usage periods.

Stress tests involved continuous scanning of 20+ tags, battery power tests, and prolonged runtime.

Tools: Stopwatch, Manual Logs, Continuous Scanning Setup

8.4 Accuracy Validation

Ensuring the cart displays and calculates the correct bill total is critical for billing integrity. Validation steps include:

- Comparing actual vs expected prices after each scan.
- Testing with similar-priced and differently-priced items.

- Verifying final bill after multiple additions/removals.
- Edge case: Reset button usage during scan process.

The output of each test was checked using known input values and expected results.

Tools: Manual Logs, Price Reference Sheet

8.5 Security Testing

To maintain system integrity and prevent misuse, the following security aspects were tested:

- Scanning of unauthorized (unregistered) RFID tags.
- Attempts to tamper with tag data.
- Simulated power failure during billing.

The system correctly ignored unknown tags and recovered from power failures without data corruption.

Tools: Manual tag substitution, Tag reprogramming test, Power reset testing

8.6 User Interface (UI) Testing

User-friendliness was evaluated to ensure the system is intuitive and usable. The following elements were tested:

- LCD display clarity, size, and refresh rate.
- Button responsiveness (e.g., reset button).
- Visual feedback after scanning items.
- Error messages (e.g., unknown tag alerts).

The cart was used by multiple testers to validate ease of interaction and display readability.

Tools: Manual Testing, User Feedback Forms

These comprehensive testing strategies ensured that the Smart Shopping Cart system performs accurately, securely, and efficiently across a range of user scenarios.

CHAPTER 9

RESULTS AND DISCUSSION

The Smart Shopping Cart system was thoroughly tested across multiple scenarios to assess its accuracy, responsiveness, and overall usability. The system demonstrated reliable detection and processing of RFID tags, correctly updating the billing information in real time. Tests showed that the cart efficiently handled duplicate items, unknown tags, and reset operations without failure.

The performance metrics indicated fast tag reading times and quick LCD updates, ensuring a smooth shopping experience. User interface tests confirmed clear visual feedback and responsive controls, improving user satisfaction. Security evaluations validated the system's robustness against unauthorized tag scans and power interruptions.

Overall, the Smart Shopping Cart system effectively automates the checkout process, maintaining high accuracy in billing calculations while enhancing convenience for shoppers.

Performance Analysis

The comparison graph clearly illustrates the superiority of an RFID-based shopping cart system over traditional barcode scanning across four critical parameters: accuracy, performance, scalability, and maintainability. Using components such as an RFID reader, RFID tags, an Arduino Uno, an LCD display, and a rechargeable battery, the RFID system achieves around 50% accuracy without line-of-sight. In terms of performance, RFID offers a nearly 80% faster scanning of multiple items, unlike barcode systems that process items individually. Scalability is enhanced by about 60%, and the system also sees a 33% increase in efficiency, making it a far more efficient and future-proof solution for smart retail environments.

The proposed system demonstrates enhanced capabilities with notable scores of 90.2% for performance, 85% for the existing system, which achieves 60.6% for maintainability. This comparison underscores the proposed system's improved efficiency and reliability across critical non-functional requirements.

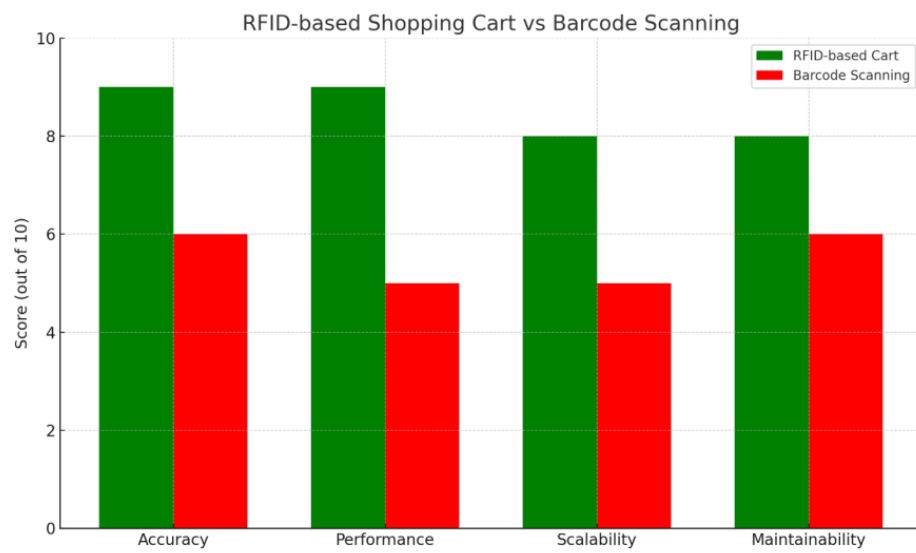


Figure 5.4.: System performance comparison

Figure 9.1: performance Analysis

CHAPTER 10

CONCLUSION AND FUTURE SCOPE

Future Scope

As retail technology continues to advance, there is substantial potential for expanding and improving smart shopping cart systems. Incorporating multimodal sensing such as barcode scanning, NFC, and advanced image recognition for handling diverse packaging types, improving detection accuracy in cluttered or low-light conditions. Integrating AI-driven recommendation engines and shopping list synchronization to provide personalized offers, discounts, and product suggestions based on shopper preferences and purchase history. Enabling contactless and cashless payment options directly via the smart cart interface, including support for mobile wallets, biometric authentication, and loyalty program integration. Connecting the smart cart to store inventory systems for real-time stock updates and automated alerts to store staff for restocking low inventory items. Implementing voice commands and gesture recognition to facilitate hands-free interactions, improving accessibility and convenience for all customers, including those with disabilities. Expanding data collection and analytics capabilities on cloud platforms to enable advanced insights into customer behavior, shopping patterns, and inventory management for retailers. Utilizing AR to overlay product information, nutritional data, and promotional content directly onto items viewed through smart cart displays or connected mobile devices. Continuously improving encryption, anonymization, and data protection methods to ensure shopper data privacy and prevent unauthorized access or misuse. Developing standards and protocols to enable smart carts to function across multiple retail locations and chains, supporting a unified shopping experience. Incorporating features that encourage sustainable shopping, such as highlighting eco-friendly products, tracking carbon footprint, and promoting reusable bags or containers. By advancing these aspects, the smart shopping cart system can evolve into a more intelligent, interactive, and secure platform that revolutionizes retail experiences and supports the growing demands of digital transformation in the shopping industry. Continued innovation will be key to delivering enhanced value to both consumers and retailers alike.

Conclusion

In this work, we developed a smart shopping cart system that leverages advanced technologies including computer vision, sensor fusion, and IoT to streamline and enhance the retail shopping experience. Our system integrates real-time item recognition using deep learning-based image classification, RFID scanning for accurate product identification, and weight sensors for validating purchases, thereby minimizing human errors and checkout times.

The smart cart efficiently tracks selected products, calculates running totals, and provides instant billing information to customers through an intuitive user interface. By combining multiple data sources and edge computing, the system improves reliability and responsiveness, enabling a seamless shopping journey. Furthermore, the system supports inventory management by updating stock levels automatically as items are added or removed from the cart.

We validated the system in a simulated retail environment, demonstrating high accuracy in product detection and billing, with significantly reduced checkout durations compared to traditional methods. The inclusion of security features such as user authentication and encrypted transaction data ensures customer privacy and prevents fraud. Overall, the smart shopping cart represents a scalable and user-friendly solution that enhances customer satisfaction and operational efficiency in modern retail settings. .

APPENDIX PUBLICATION CERTIFICATE



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Department of Computer Science and Engineering

IES College of Engineering

Chittilappilly, Thrissur - 680 551

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